

ZOMATO DATA ANALYSIS

Phase 8 — Advanced Business Analysis Report

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41,410
Restaurants Analyzed

72
Locations Scored

$R^2 = 0.82$
ML Accuracy

95.6/100
Top Expansion Score

EXECUTIVE SUMMARY

This report presents advanced business analysis on 41,410 Bengaluru restaurants using a custom Expansion Score algorithm and Machine Learning rating prediction model. The analysis was conducted as Phase 8 of a complete end-to-end data analytics project covering ETL, EDA, SQL analysis, Power BI dashboarding, and ML modelling.

EXPANSION ANALYSIS

Methodology

$\text{Expansion Score} = (40\% \times \text{Demand Pressure}) + (35\% \times \text{Avg Rating}) + (25\% \times \text{Supply Gap})$

- All metrics normalized to 0–100 using MinMaxScaler (scikit-learn)
- Only locations with 30+ restaurants included (statistically significant)
- Total locations analyzed: 72
- Critical priority zones identified: 3

Top 10 Expansion Targets

Rank	Location	Score	Priority	Restaurants	Avg Rating
1	Lavelle Road	95.6	CRITICAL	481	4.14
2	Church Street	91.1	CRITICAL	546	3.99
3	St. Marks Road	81.6	CRITICAL	343	4.02
4	Koramangala 3rd Block	78.1	HIGH	191	4.02
5	Koramangala 4th Block	76.2	HIGH	841	3.92
6	Koramangala 5th Block	75.8	HIGH	2,297	4.01
7	Cunningham Road	69.9	HIGH	475	3.90
8	MG Road	63.5	HIGH	793	3.85
9	Residency Road	62.9	HIGH	604	3.86
10	Indiranagar	60.5	HIGH	1,803	3.83

■ *Business Recommendation: Zomato should prioritize restaurant onboarding in Lavelle Road and Church Street before these markets reach saturation. Both score above 90/100 on the Expansion Index.*

RATING PREDICTION MODEL

Model Configuration

Parameter	Value
Algorithm	Random Forest Regressor
n_estimators	100
Train / Test split	80% / 20% (33,128 / 8,282 restaurants)
Random state	42
Features used	5 (votes, cost_for_two, rest_type, online_order, book_table)

Model Performance Comparison

Model	R ² Score	RMSE	Verdict
Linear Regression	0.2920	0.3696	Weak — linear assumption insufficient
Random Forest	0.8157	0.1885	WINNER — 179% better R² score

Feature Importance Rankings

Rank	Feature	Importance	Business Interpretation
1	votes (customer engagement)	69.3%	DOMINANT — 4.7x more important than price
2	cost_for_two	14.7%	Price tier has moderate influence on quality
3	rest_type (restaurant format)	10.9%	Format matters — microbreweries rate highest
4	online_order	3.0%	Minimal direct impact on customer rating
5	book_table	2.1%	Low weight but confirms premium segment signal

■ *Key Finding: Customer engagement (votes) is 4.7x more important than price in predicting restaurant quality. Zomato should weight engagement metrics higher in their restaurant ranking algorithm.*

BUSINESS RECOMMENDATIONS

1. EXPAND in 3 Critical Priority Zones

Lavelle Road (95.6), Church Street (91.1), and St. Marks Road (81.6) show the highest Expansion Score. These locations have high demand pressure, strong existing ratings, and relatively low restaurant supply. Onboard new restaurant partners here before saturation.

2. REDESIGN the Rating Algorithm

Customer votes drive 69% of rating prediction — 4.7x more than cost. Zomato's current recommendation engine likely underweights engagement signals. Reweighting votes in the ranking algorithm would improve recommendation quality significantly.

3. TARGET the Premium Cuisine Gap

Modern Indian (4.31 avg), European (4.26), and Mediterranean (4.20) cuisines consistently outperform in ratings but have very low restaurant counts. Partner with premium chefs in these categories — high demand exists with low competition.

4. SCALE the Table Booking Program

Table booking restaurants rate 0.52 points higher and charge 2.6x more (Rs.1,276 vs Rs.482 avg cost). Expanding Zomato Gold/Pro to incentivize more restaurants to enable table booking would capture significant revenue upside.

5. QUALITY IMPROVEMENT for Bottom 10 Locations

Bommanahalli (3.19 avg), RT Nagar (3.46), and other peripheral areas have high restaurant counts but poor quality scores. Targeted restaurant training and quality certification programs in these areas would improve overall platform rating.

FILES GENERATED

File	Description
reports/expansion_score.png	Top 15 expansion targets bar chart + priority pie chart
reports/feature_importance.png	What drives restaurant ratings — feature importance chart
reports/chart1_rating_distribution.png	Rating distribution histogram + category bars
reports/chart2_table_booking.png	Table booking vs no booking (rating + cost comparison)
reports/chart3_cost_vs_rating.png	Cost bracket vs avg rating gradient chart
reports/chart4_location_rating.png	Top 10 vs Bottom 10 locations horizontal bar charts
reports/chart5_cuisine_analysis.png	Supply vs quality side-by-side cuisine analysis
reports/Phase8_Business_Report.pdf	This report (PDF version)